Events

Query and transaction events as a JSON stream.

Install & load

```
INSTALL events FROM community;
LOAD events;
```

01 At a glance

Forward DuckDB lifecycle events to an executable handler as JSON lines on standard input. Use the stream for operational logs, query timing and transaction traces, with a session tag to identify each running process.

DuckDB event → JSON line → Handler stdin

02 Four settings & event keys

`events_destination`

Absolute path to the executable that consumes JSON lines.

`events_types`

List of enabled query, connection or transaction event types.

`events_session_name`

Tag each event with a process or workload identifier.

`events_async`

TRUE enables asynchronous delivery; default is FALSE.

`query_id / transaction_id`

Correlate query pairs and the enclosing transaction.

`has_error / error_message`

Inspect failures on query_end and transaction_rollback events.

03 Keep in mind

> **A handler is required**
>
> Create an executable with a shebang at the configured absolute path. It must read stdin line by line and persist or process each JSON object. The extension launches the handler with DuckDB's privileges.

04 Capture query activity

MANUAL SETUP

```
SET events_session_name = 'analytics';
SET events_types = [
  'query_begin', 'query_end'
];
SET events_async = TRUE;
SET events_destination =
  '/usr/local/bin/query-events';
SELECT sum(i) FROM range(1000) t(i);
```

Requires an installed compatible Events build and your executable JSONL handler.

05 Trace transaction boundaries

MANUAL SETUP

```
SET events_types = [
  'transaction_begin',
  'transaction_commit',
  'transaction_rollback'
];
BEGIN;
SELECT 42;
COMMIT;
```

With the handler running, group transaction events by transaction_id.

06 In practice

> **Lifecycle and availability**
>
> Settings belong to the running process; reapply them at startup. Async delivery trades reliability for query latency. Persist events in the handler when durability matters. Community package availability varies by DuckDB version and platform.

Full docs & examples ↗ Source ↗